

Egg-laying mammals

PHYLUM	Chordata
CLASS	Mammalia
ORDER	Monotremata
FAMILIES	2
SPECIES	5

Also known as **monotremes**, this group comprises the duck-billed platypus and the echidnas. Monotremes are the only mammals that lay eggs. The duck-billed platypus has a ducklike bill, a beaverlike tail, and webbed feet. Echidnas have a tubular beak, huge claws for digging, and spines. The duck-billed platypus lives in freshwater in eastern Australia; echidnas exist in varied habitats in Australia and New Guinea.



PLATYPUS



ECHIDNA

DIFFERENT SNOUTS

The platypus (top) has a flattened bill, covered with sensitive skin. Echidnas, however, have a cylindrical beak. In both, the snout is used to probe for and locate food.

Anatomy

Monotremes are short-legged animals with a small head and tiny eyes. In all species, the digestive, urinary, and reproductive tracts empty into a common chamber (the cloaca), which terminates in a single exit ("monotreme" means "one hole").

webbing between digits



WEBBED FEET

In water, the duck-billed platypus is propelled by powerful thrusts of its fully webbed front feet.

STICKY CATCH

The short-nosed echidna uses its long, slender tongue to catch termites and ants. A sticky coating of saliva ensures that the insects do not escape.

Feeding

The platypus uses its sensitive bill to probe the beds of rivers or lakes for crustaceans and insect larvae. The short-nosed echidna digs with its powerful claws for ants, termites, or earthworms; long-nosed echidnas mainly eat earthworms. Adult monotremes lack teeth: food is ground between plates or spines in the mouth.

AQUATIC LIFE

The duck-billed platypus is a strong swimmer. Its coat is waterproof and has a dense underfur for insulation. When diving, the eyes and ears are closed.



Tachyglossus aculeatus

Short-beaked echidna



Length 12–18 in (30–45 cm)

Tail ¾ in (1 cm)

Weight 5½–15 lb (2.5–7 kg)

Location Australia (including Tasmania), New Guinea

Social unit Individual

Status Least concern



The spines of this species, also called the spiny anteater, are longer than the fur between them. Active both day and night, this echidna is solitary and can become torpid in very cold or hot weather, when its temperature falls from the normal 88–92° F (31–33° C) to as low as 39° F (4° C). It eats a variety of ants, termites, grubs, and worms. These are detected by smell and perhaps by sensors on the long snout that detect electric signals. The head is small and there is no discernible neck.



Zaglossus bartoni

Eastern long-beaked echidna



Length 19–27 in (48–68 cm)

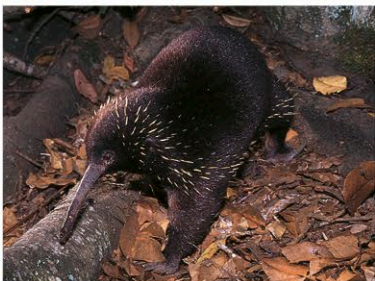
Tail None

Weight 8¾–14 lb (4.2–6.5 kg)

Location New Guinea

Social unit Individual

Status Vulnerable



The downcurved snout of this species may exceed 8 in (20 cm) in length, with a tiny mouth at the tip and small, close-set eyes at its base. One of the largest monotremes, it is slow-moving, rolling into a spiny ball for defense. The spines are often only just visible through its long black fur. The female lays 4–6 eggs into her pouch in July. Hatching occurs 10 days later, and the infants remain in the pouch for 6–7 weeks, or until their spines develop.

Ornithorhynchus anatinus

Duck-billed platypus



Length 16–23½ in (40–60 cm)

Tail 3¼–6 in (8.5–15 cm)

Weight 1¼–5½ lb (0.8–2.5 kg)

Location E. Australia, Tasmania

Social unit Individual

Status Near threatened



The ducklike, beak-shaped mouth, sprawling, reptile-type gait, and flattened, almost scaly, beaverlike tail make the duck-billed platypus an unmistakable animal. Its waterproof body fur has a plush texture that is reminiscent of that of a mole. The color of the body is dark brown to reddish brown on top and pale

brown to silvery cream underneath. The probing bill is very sensitive to both touch and water-borne electrical signals from the muscles of its small aquatic prey. It has an unusually high density of red blood cells to enable it to make deep dives in pursuit of its quarry. Platypuses are solitary, yet occupy overlapping home ranges. However, males in the breeding season defend these territories. The male has a venomous spur on each hind foot, with which he attempts to wound his rivals. Home is a bankside burrow, usually about 16 ft (5 m) long but it may reach up to 98 ft (30 m). After a gestation period of one month, the female incubates her clutch of 1–3 soft, leathery-shelled eggs here for 10 days. After the young hatch, she suckles them for 4 months in the burrow, leaving them walled in for periods of up to 38 hours while she forages.



partially webbed back feet

fully webbed front feet